

FREDERICTON AQUATIC CENTRE, NB, Canada

WMO: 716445

Lat: 45.966N

Lon: 66.651W

Elev: 8

StdP: 101.23

Time Zone: -4.00 (NAA)

Period: 05-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.5	-19.4	-26.5	0.3	-21.3	-23.9	0.4	-18.8	12.4	-8.4	11.4	-8.8	2.4	280	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	30.4	21.9	28.7	21.1	27.3	20.0	23.4	28.2	22.5	26.7	21.6	25.3	3.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	16.6	25.5	21.1	15.8	24.7	20.3	15.0	23.9	69.5	28.6	66.3	27.0	63.1	25.2	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.2	8.0	DB	-26.2	33.5	4.4	1.3	-29.4	34.4	-32.0	35.1	-34.4	35.8	-37.6	36.8	(4)
(5)			WB	-26.4	25.1	4.2	1.0	-29.5	25.9	-32.0	26.5	-34.4	27.1	-37.4	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-7.1	-6.8	-1.0	6.3	12.6	17.7	21.6	20.7	16.6	10.1	4.0	-2.9
	DBStd	10.99	6.92	5.94	5.56	3.73	3.75	3.10	2.53	2.74	3.73	3.95	4.62	5.48
	HDD10.0	2117	530	471	343	122	14	0	0	0	1	48	188	401
	HDD18.3	4118	788	704	599	362	182	47	4	9	76	256	430	659
	CDD10.0	1288	0	0	1	10	96	233	361	331	199	51	7	0
	CDD18.3	248	0	0	0	0	5	29	106	82	25	1	0	0
	CDH23.3	1752	0	0	1	6	78	235	741	528	155	9	0	0
	CDH26.7	442	0	0	0	1	20	52	200	129	41	1	0	0
(14) Wind	WSAvg	3.1	3.6	3.7	3.8	3.4	3.0	2.7	2.3	2.4	2.6	3.1	3.1	3.6
(15) Precipitation	PrecAvg	1140	87	74	91	84	95	95	99	82	100	114	108	111
	PrecMax	1499	152	132	151	147	192	182	204	182	229	259	177	192
	PrecMin	722	49	27	32	27	22	44	27	18	21	32	45	40
	PrecStd	156	27	26	31	32	37	35	52	44	53	58	34	40
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	8.8	17.6	22.4	29.2	30.5	32.5	31.9	31.1	24.6	18.5	13.2
		MCWB	11.1	6.8	11.1	13.4	19.7	20.4	23.0	23.2	23.3	18.0	17.1	11.9
	2%	DB	7.8	4.9	11.1	18.5	25.3	28.2	30.5	29.8	27.1	20.5	15.8	9.1
		MCWB	6.7	2.6	6.7	10.8	16.2	20.0	22.1	22.1	20.6	15.7	13.6	8.3
	5%	DB	4.4	3.0	8.5	15.5	22.4	26.0	28.8	27.9	24.7	18.2	13.7	6.6
		MCWB	3.1	1.5	4.7	9.6	14.7	18.7	21.4	20.7	19.5	15.5	11.8	5.6
	10%	DB	2.4	1.9	6.3	12.9	19.7	24.1	27.4	26.3	22.8	16.3	11.3	4.6
		MCWB	1.4	0.8	3.3	7.9	13.4	17.4	20.4	20.1	18.4	13.5	9.5	3.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	7.5	12.6	16.0	20.8	22.7	24.7	24.3	23.7	19.7	17.4	11.9
		MCDB	12.1	8.6	17.0	20.6	26.6	27.8	30.2	29.3	29.7	22.3	18.4	13.0
	2%	WB	6.8	2.9	7.7	12.9	18.0	21.5	23.5	23.2	22.1	17.5	14.5	8.2
		MCDB	7.6	4.1	10.1	16.1	22.6	25.8	28.3	27.9	25.0	19.3	15.2	8.7
	5%	WB	3.2	1.9	5.3	10.4	16.4	20.0	22.6	22.3	20.8	16.0	12.2	5.8
		MCDB	4.1	2.8	7.8	13.9	20.4	23.4	26.9	26.3	23.3	18.7	13.5	6.3
	10%	WB	1.4	0.9	3.6	8.5	14.6	18.8	21.7	21.3	19.4	14.0	9.5	3.5
		MCDB	2.3	1.9	5.8	12.2	18.2	22.3	25.5	24.6	22.2	15.5	10.9	4.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.9	10.0	9.8	10.3	9.8	9.9	9.7	10.0	8.1	7.5	7.3
		MCDBR	10.8	9.6	12.5	15.5	15.7	13.7	12.2	12.2	11.9	11.1	11.1	9.2
	5% WB	MCWBR	10.4	7.8	8.6	8.9	7.9	6.5	5.5	5.6	6.4	7.5	9.2	8.9
		MCWBR	11.1	8.3	11.7	12.9	12.8	11.5	10.4	10.4	10.1	9.8	10.0	9.4
(40) Clear-Sky Solar Irradiance	taub		0.275	0.286	0.318	0.346	0.374	0.406	0.425	0.400	0.371	0.328	0.306	0.287
		taud	2.455	2.401	2.425	2.424	2.388	2.305	2.272	2.344	2.400	2.525	2.538	2.475
	Ebn at Noon		843	899	914	913	893	862	839	848	843	841	799	785
		Edh at Noon	67	86	98	108	116	127	129	116	100	77	63	60
(44) All-Sky Solar Radiation	RadAvg		1.49	2.36	3.51	4.36	4.99	5.40	5.48	4.85	3.92	2.41	1.41	1.10
	RadStd		0.16	0.21	0.23	0.41	0.57	0.41	0.43	0.31	0.29	0.23	0.16	0.10

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+50	N/A

Nomenclature: See separate page